Undergraduate studies

Mathematical Finance (Bachelor of Mathematics - Honours)

Systems of Study

Co-operative
Regular

Minimum Average(s) Required

- A minimum cumulative overall average of 60.0%.
- A minimum cumulative major average of 70.0%: all math courses.

Graduation Requirements

- See Bachelor of Mathematics degree-level requirements.
- Complete all the required courses listed below.
- Complete a minimum of 13.0 units of math courses.
- Complete a minimum of 5.0 units of non-math courses.

Co-operative Education Program Requirements

For students in the co-operative system of study, see Bachelor of Mathematics co-operative education program requirements.

Course Requirements

Required Courses

- Complete all of the following
  - Complete all the following:
    - ACTSC231 - Introductory Financial Mathematics (0.50)
    - ACTSC372 - Investment Science and Corporate Finance (0.50)
    - ACTSC445 - Quantitative Enterprise Risk Management (0.50)
    - ACTSC446 - Mathematics of Financial Markets (0.50)
    - PMATH351 - Real Analysis (0.50)
    - PMATH450 - Measure Theory (0.50)
    - STAT330 - Mathematical Statistics (0.50)
    - STAT331 - Applied Linear Models (0.50)
    - STAT333 - Stochastic Processes 1 (0.50)
    - STAT443 - Forecasting (0.50)

- Complete 1 of the following
  - Complete all the following:
    - AFM101 - Introduction to Financial Accounting (0.50)
  - Complete all the following: BUS127W

- Complete 1 of the following
  - Complete all the following:
    - AFM102 - Introduction to Managerial Accounting (0.50)
  - Complete all the following: BUS247W

- Complete 1 of the following
  - Complete 1 of the following:
    - AFM131 - Introduction to Business in North America (0.50)
    - ARBUS101 - Introduction to Business in North America (0.50)
  - Complete all the following: BUS111W
- Complete 1 of the following:
  - AMATH242 - Introduction to Computational Mathematics (0.50)
  - CS335 - Computational Methods in Business and Finance (0.50)
  - CS371 - Introduction to Computational Mathematics (0.50)
- Complete 1 of the following:
  - AMATH250 - Introduction to Differential Equations (0.50)
  - AMATH251 - Introduction to Differential Equations (Advanced Level) (0.50)
  - AMATH350 - Differential Equations for Business and Economics (0.50)
- Complete 1 of the following:
  - AMATH351 - Ordinary Differential Equations (0.50)
  - CO250 - Introduction to Optimization (0.50)
  - CO255 - Introduction to Optimization (Advanced Level) (0.50)
  - PMATH352 - Complex Analysis (0.50)
- Complete 1 of the following:
  - ACTSC447 - Numerical Computation for Financial Modelling (0.50)
  - AMATH353 - Partial Differential Equations 1 (0.50)
  - CO372 - Portfolio Optimization Models (0.50)
  - CS476 - Numerical Computation for Financial Modelling (0.50)
  - PMATH453 - Functional Analysis (0.50)

- Complete 1 of the following
  - Complete all the following:
    - ECON101 - Introduction to Microeconomics (0.50)
  - Complete all the following: ECON120W

- Complete 1 of the following
  - Complete all the following:
    - ECON102 - Introduction to Macroeconomics (0.50)
  - Complete all the following: ECON140W

- Complete 1 of the following
  - Complete all the following:
    - ECON201 - Microeconomic Theory for Business and Policy (0.50)
  - Complete all the following: ECON260W

- Complete 1 of the following
  - Complete all the following:
    - MATH247 - Calculus 3 (Advanced Level) (0.50)
  - Complete all the following:
    - MATH237 - Calculus 3 for Honours Mathematics (0.50)
    - PMATH333 - Introduction to Real Analysis (0.50)

Additional Constraints

1. Students may only complete one course from any cross-listed set.
2. Students enrolled in the Business Administration and Mathematics double degree academic plan:
   1. May substitute BUS393W for ACTSC372.
   2. Are exempt from taking STAT443, but can count it under the "Complete 1 of AMATH353, CO372, CS476, PMATH453" requirement if they choose to take it.
3. Students currently or previously enrolled in the following academic plans (Business Administration and Mathematics double degree, Mathematics/Business Administration, Mathematics/Financial Analysis and Risk Management, Information Technology Management, and Mathematical Optimization – Business Specialization) may substitute:
   1. STAT371 for STAT331.

Offered by Faculty(ies)

Faculty of Mathematics



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The University of Waterloo acknowledges that much of our work takes place on the traditional territory of the Neutral, Anishinaabeg, and Haudenosaunee peoples. Our main campus is situated on the Haldimand Tract, the land granted to the Six Nations that includes six miles on each side of the Grand River. Our active work toward reconciliation takes place across our campuses through research, learning, teaching, and community building, and is co-ordinated within the Office of Indigenous Relations <https://uwaterloo.ca/indigenous>.

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